

Supplementary Material

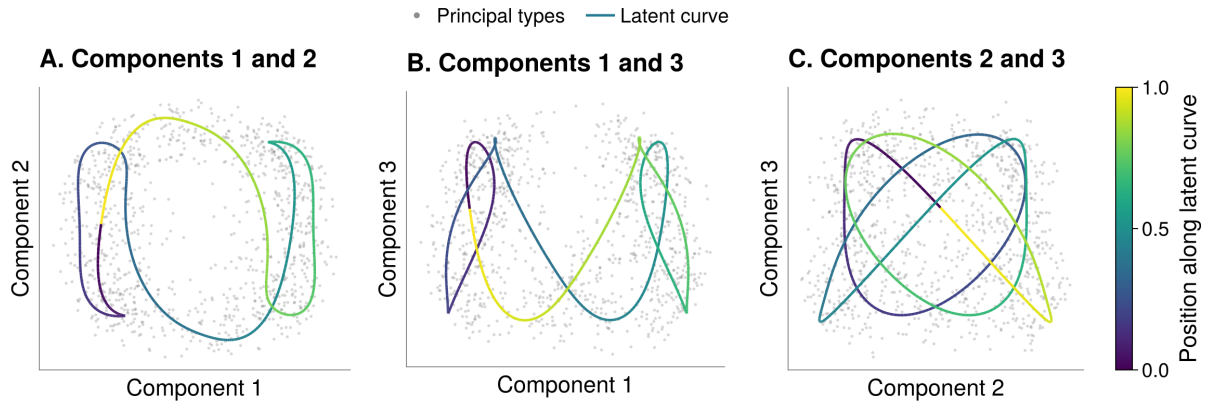


Figure S1: Principal types and their latent curve. All 1000 baseline types from realization 1, projected onto pairs of the curve's first three principal components (A–C). Gray points are realized types; the line shows the latent curve, colored by position along it. Matching and learning use the full eight-dimensional types, not these projections.

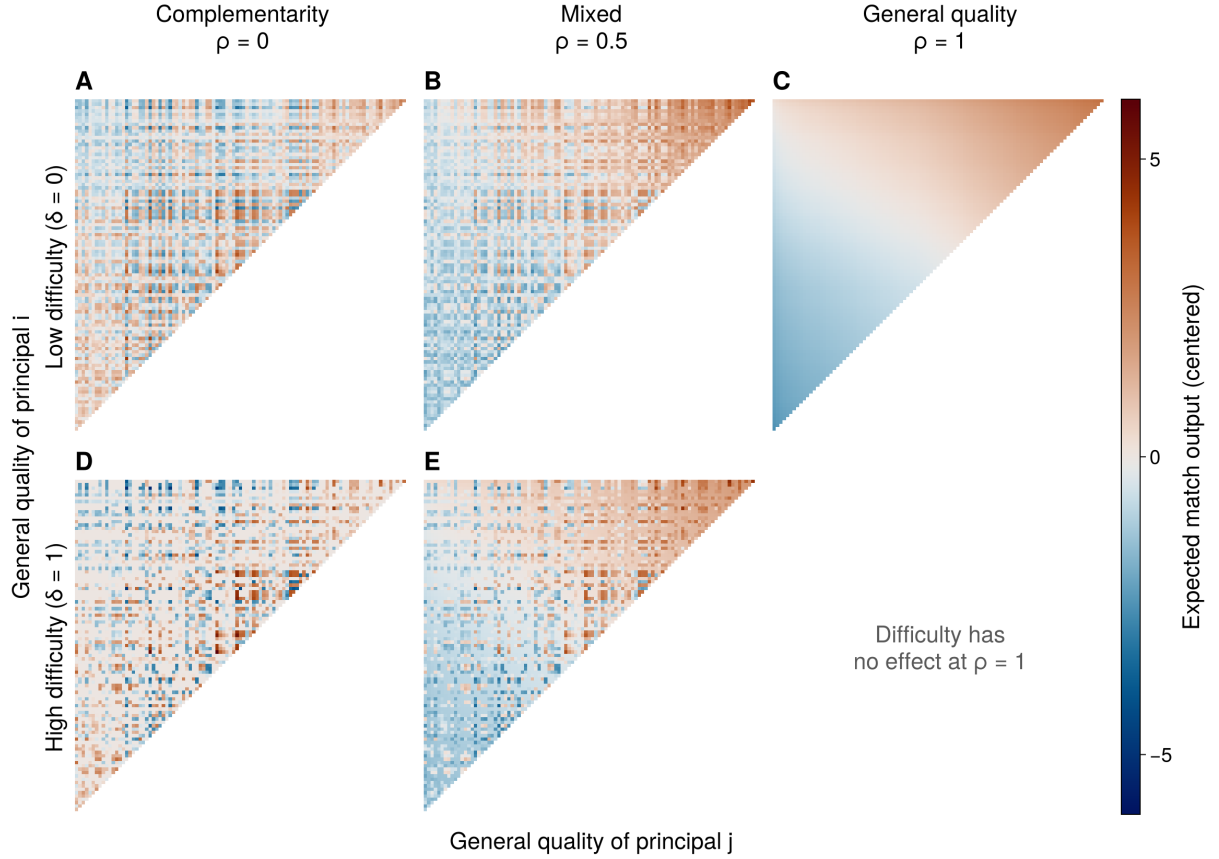


Figure S2: Expected match output across pairs of principal types. 100 types selected at evenly spaced general-quality ranks from realization 1 (1000 principals). Columns vary general-quality share ρ ; rows vary difficulty δ . The $\rho = 1$ condition appears once because difficulty is inactive. Types and matching-function objects c , A , and B are fixed across panels. General quality increases rightward and upward. Only the upper triangle of each symmetric matrix is shown, excluding self-matches. A shared color scale shows expected output minus the grand mean, without match noise.

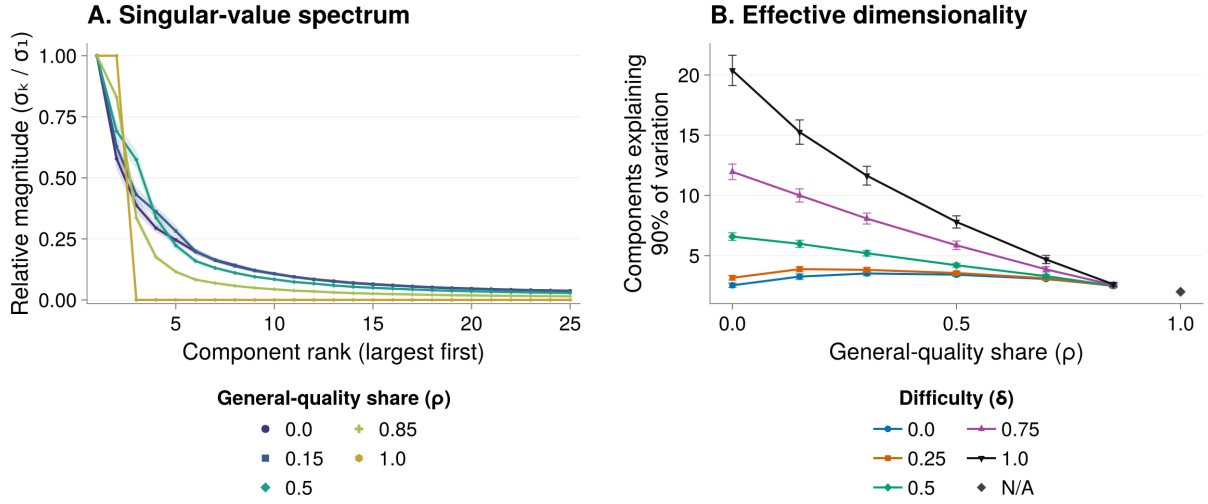


Figure S3: Dimensionality of expected match output. Centered conditional-output matrices, without match noise, summarized over 50 independent realizations. Types and matching-function objects c , A , and B are fixed within each realization as ρ and δ vary. A: the first 25 singular values at $\delta = 0.5$, divided by the largest within each realization. B: the fewest singular components accounting for 90% of the sum of squared singular values within each realization. Lines and markers are means; ribbons and whiskers are pointwise 95% Monte Carlo intervals. Colors and symbols identify general-quality share ρ in A and difficulty δ in B. N/A denotes the unconnected gray diamond at $\rho = 1$: difficulty affects only complementarity and is not applicable under pure general quality.

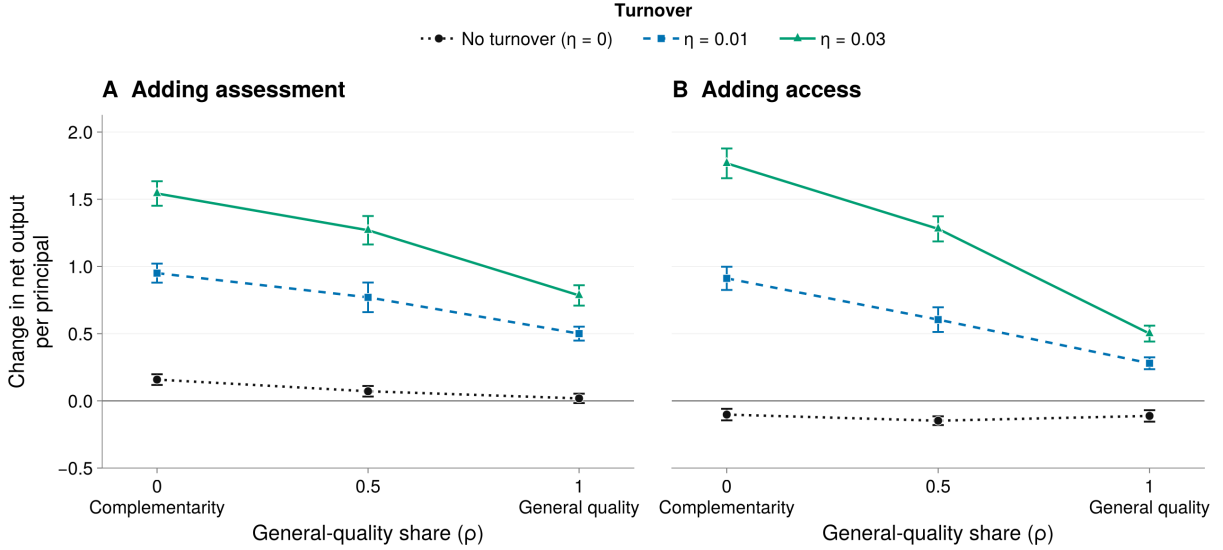


Figure S4: Contributions of assessment and access across matching composition. A: full service minus access-only. B: full service minus assessment-only. Both panels show changes in net output per principal. The 9 regimes combine 3 matching compositions with turnover rates of 0%, 1%, and 3% per period; other parameters remain at baseline. Difficulty is inapplicable under pure general quality ($\rho = 1$). Points show late-mean differences paired by random seed, using 20 seeds per service version and regime. Bars show pointwise 95% Monte Carlo intervals. Positive values indicate higher net output under full service; connecting lines are visual guides.

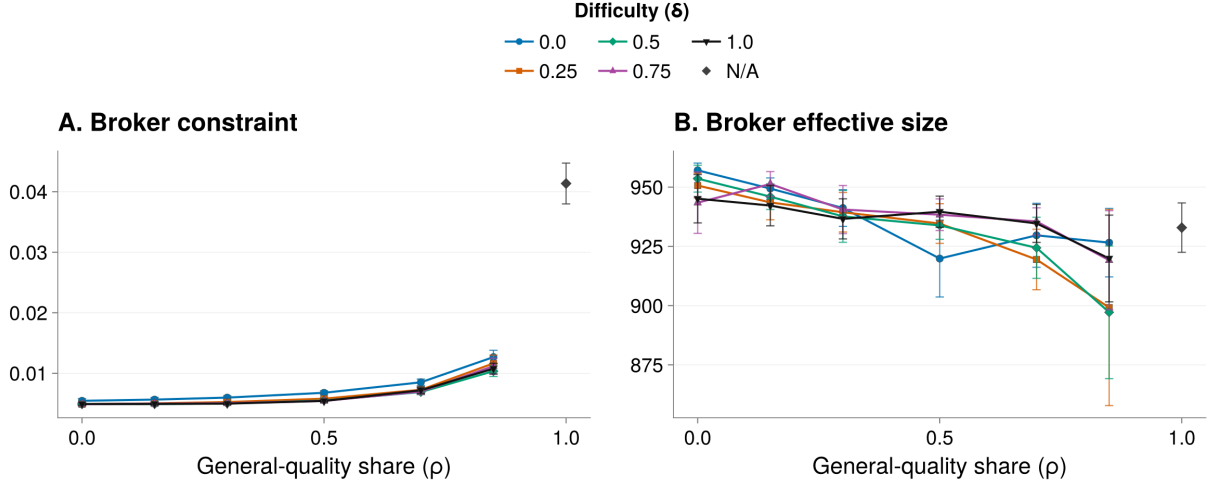


Figure S5: Alternative structural measures across the matching problem. A: Burt constraint, the concentration of the broker's direct and indirect ties. B: effective size, which discounts broker contacts connected to one another. Points are late-window means across the $\rho \times \delta$ design, with other parameters at baseline. Colors and symbols identify difficulty δ . N/A denotes the unconnected gray diamond at $\rho = 1$: difficulty affects only complementarity and is not applicable under pure general quality. Whiskers are 95% Monte Carlo intervals.

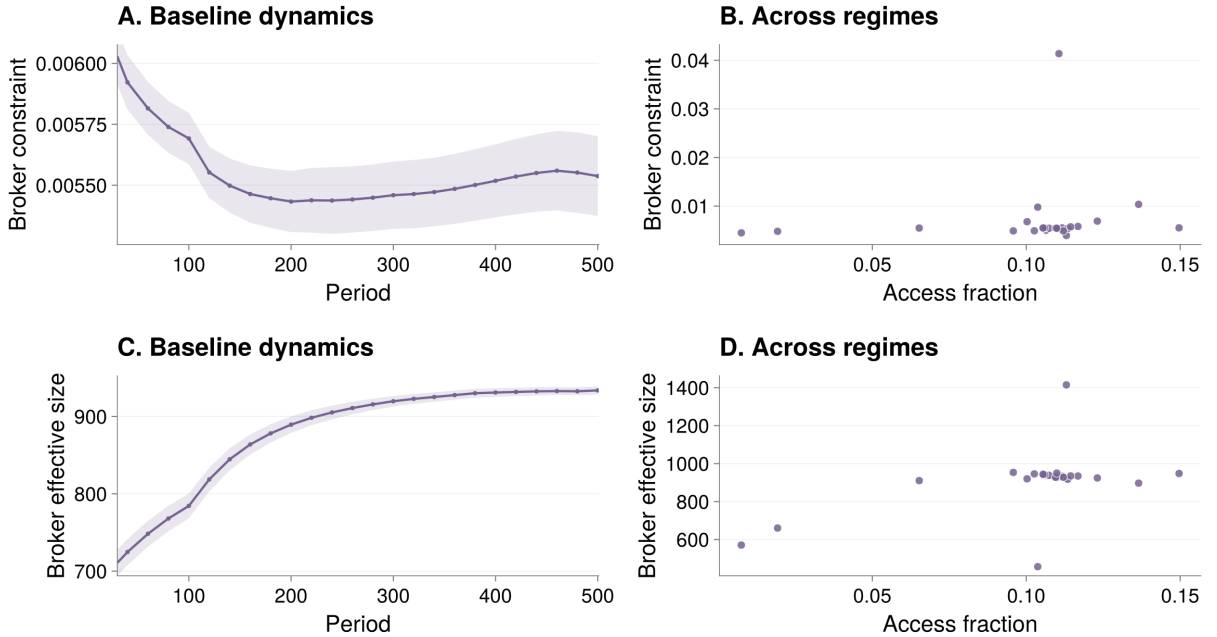


Figure S6: Alternative structural measures over time and against access. Top: the broker's Burt constraint. Bottom: the broker's effective size. A and C: baseline dynamics, recorded every 20 periods and smoothed within each seed using a 5-observation rolling mean. Lines are ensemble means; ribbons are pointwise 95% Monte Carlo intervals. Time starts at $t = 30$. B and D: late-window means for the 26 regimes that vary one parameter at a time from baseline. Vertical scales differ between baseline dynamics and cross-regime comparisons.

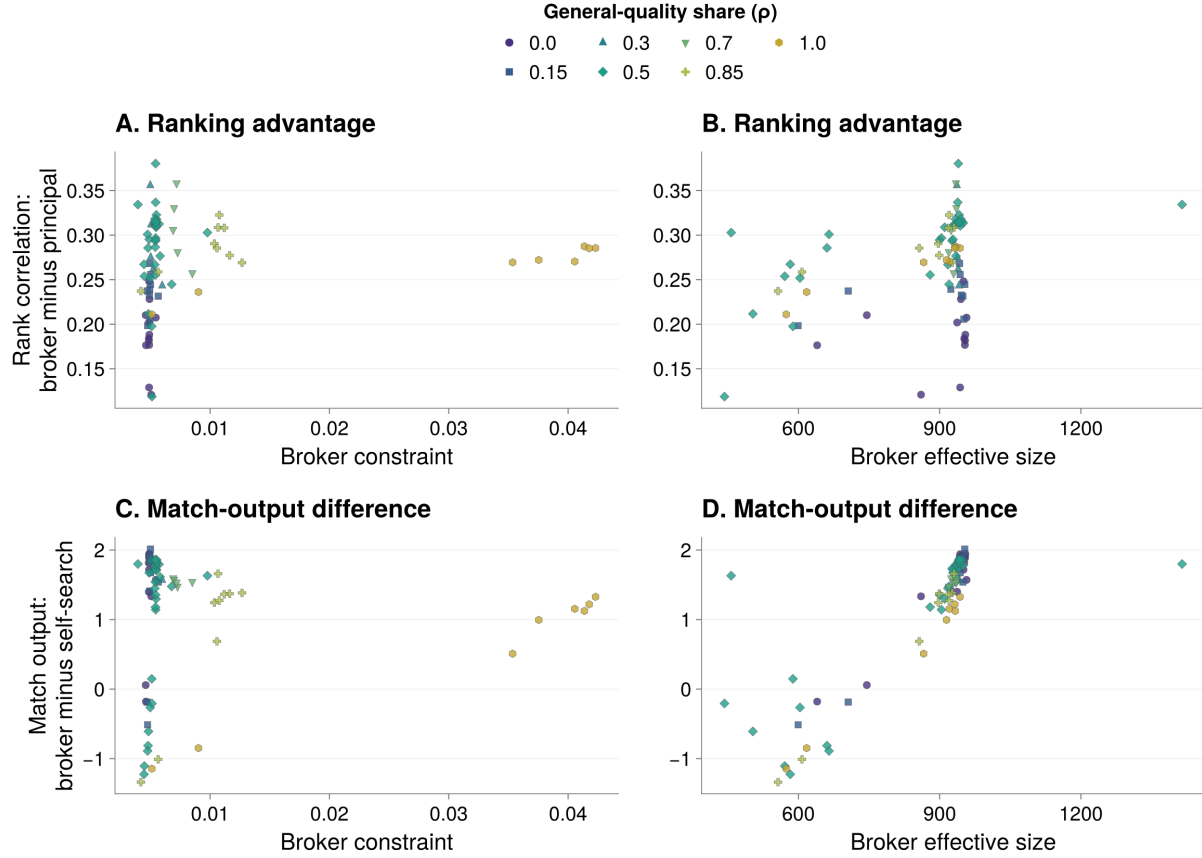


Figure S7: Ranking and output differences by alternative structural measures. A–B: ranking advantage, broker minus principal. C–D: mean realized output per completed match, broker minus self-search, before costs and fees. Horizontal axes show the broker’s Burt constraint (A, C) or effective size (B, D). Each point is a late-window mean for one of the 80 tested regimes. Colors and symbols identify general-quality share ρ .